



Synbiotic alginate microcapsules incorporating a date seed polysaccharide–stabilized Pickering emulsion improve probiotic gastrointestinal survival and modulate SCFA production, gut microbiota, and fecal metabolome

Athira Jayasree Subhash^a, Gafar Bamigbade Babatunde^a,
Maduni Jayahansi Paththuwe Arachchi^a, Abdelmoneim Ali^a, Afaf Kamal-Eldin^a,
Mutamed Ayyash^{a,*}

^a Department of Food Science, College of Agriculture and Veterinary Medicine, United Arab Emirates University (UAEU), Al-Ain, United Arab Emirates

ARTICLE INFO

Keywords:

Microencapsulation
Probiotics
Pickering emulsions
Polysaccharides
Emulsion
Prebiotics
Un-targeted metabolomics

ABSTRACT

This study formulated alginate microbeads incorporating a date seed polysaccharide-stabilized pickering emulsion (Alg:PSE) aimed at enhancing probiotic survival and regulating gut microbiota in a strain-dependent manner. *Lactocaseibacillus rhamnosus* and *Bifidobacterium longum* subsp. *longum* strains were individually encapsulated within alginate beads, alginate with the ultrasound-extracted date seed polysaccharide (UPS), and alginate with the UPS-stabilized pickering emulsion (PSE). Among various compositions evaluated herein, Alg:PSE2 was identified as the best one based on the highest encapsulation efficacy and viability rates of *L. rhamnosus* (AET3) and *B. longum* (AET8). In comparison with alginate beads, both AET3 and AET8 resulted in denser, harder, less swellable, and heat-stable matrices with smooth and less porous surfaces. Moreover, both microcapsules could significantly enhance the biological activity of bioaccessible fractions following in vitro digestion in terms of their antioxidant capacity and inhibition against α -amylase and α -glucosidase enzymes. Both AET3 and AET8 could stimulate an earlier response of fermentation within 24 h of fecal fermentation with the notable production of propionic and butyric acids. Notably, such treatment caused distinct alterations in microbial community profile by enriching short-chain fatty acids (SCFAs)-associated genera and eliminating the presence of pathogenic genera like *Klebsiella*. Unbiased metabolomics analysis using LC-MS illustrated distinctive metabolome profiles associated with the predominance of SCFAs, amino acids, and lipids.

1. Introduction

One of the critical challenges in delivering probiotics is maintaining sufficient viability of the bacteria through all stages of processing, preservation, and gastric transit to achieve any significant effect on the colon. Despite being well-known for their ability to alter the composition of gut microbiota, provide support for the intestinal barrier, regulate immune system reactions, and improve gastrointestinal function, the efficiency of these strains depends on their identity and resistance to gastric and bile stress [1–7]. While the optimal viable count of about 10^6 – 10^7 CFU/g or ml is often cited, probiotic colonies typically reduce significantly before arriving in the colon as a result of acidity, digestive enzymes, oxygen, humidity, and thermal stress [5,8,9]. Therefore,

improving probiotic survivability and achieving a controlled release in the colon is a primary goal of formulation development.

Microencapsulation is currently considered among the most efficient methods of protecting probiotics from all the above-mentioned stresses [8,10]. Being one of the most widespread materials for encapsulation, alginate gelled using Ca ions demonstrates moderate crosslinking, excellent biocompatibility, and the ability to retain living cells [11]. Unfortunately, alginate microspheres often reveal high porosity, weak mechanical stability, and poor resistance to digestive juices that may limit their protective capacity. To address these issues, recent approaches focus on the reinforcement of alginate with various polysaccharides, proteins, lipids, and emulsion systems to increase the density of the polymer matrix, its barrier characteristics, and the

* Corresponding author.

E-mail address: mutamed.ayyash@uaeu.ac.ae (M. Ayyash).

<https://doi.org/10.1016/j.ijbiomac.2026.153313>

Received 27 April 2026; Received in revised form 24 June 2026; Accepted 29 June 2026

Available online 2 July 2026

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regulation of bacterial release [3,9,12–15]. Recent advances in probiotic encapsulation have moved beyond single-polymer matrices towards composite and multilayer systems—for example, layer-by-layer assembly using milk protein–polysaccharide conjugates and chitosan, and emulsion- or polysaccharide-reinforced gels prepared by extrusion, emulsification, spray drying, and ionic gelation—which collectively improve probiotic protection and enable more controlled release [10,11].

Emulsion-based formulations are especially advantageous since the presence of lipids provides an additional protection layer for the bacteria against acidic pH, bile salts, and oxygen. Still, conventional emulsions might lack stability and show an uncontrollable release after exposure to gastrointestinal enzymes [8,12]. Pickering emulsions containing solid particles adsorbed onto the oil-water interface have greater physical stability and can be prepared using natural and edible ingredients [10,19]. The incorporation of pickering emulsions into hydrogels can, therefore, result in enhanced structural integrity and probiotic protection with a controlled release.

Simultaneously, prebiotic polysaccharides from natural sources have gained attention as fermentable substrates for the growth of beneficial gut microbiota in addition to being used as functional fillers in encapsulation platforms [14,15]. In our previous research, date seed polysaccharides extracted using ultrasonic or microwave-assisted deep eutectic solvents were observed to be highly prebiotic, stimulating genera such as *Bifidobacterium*, *Ruminococcus*, and *Blautia* and enhancing short-chain fatty acids (SCFAs) production in fecal fermentation [16,17]. The purified ultrasonicated extract (UPS) was shown to possess good water-solubility and water/oil-binding capabilities and hence can act as an active polysaccharide as well as structural hydrocolloid. Such properties make UPS an excellent choice for reinforcing the alginate gel and creating a stabilized oil-in-water pickering emulsion. Hence, such a combination would create a synbiotic carrier where the matrix itself could modulate the gut microbiota.

On the other hand, there is a need for investigating the potential use of DES-derived date seed polysaccharides as co-reinforcing agents in the preparation of alginate-based pickering emulsion microcapsules for probiotics delivery. The effect of DES-derived date seed polysaccharides on the bead structures, probiotic viability, post-digestion bioactivity, and the response of downstream fecal microbiota/metabolomes has never been explored before. The majority of prior research works have only addressed the survival rates or the prebiotic effect without assessing both.

Thus, this study aimed to develop synbiotic alginate microbeads reinforced with a UPS-stabilized oil-in-water pickering emulsion (Alg:PSE) for the separate encapsulation of *Lactocaseibacillus rhamnosus* and *Bifidobacterium longum* subsp. *longum*. The study was designed to determine whether incorporating the UPS-based pickering emulsion into alginate could improve probiotic entrapment, strengthen the microbead structure, and enhance probiotic protection during gastrointestinal transit. Specifically, the objectives were: (i) to compare different Alg:PSE formulations and identify the optimal system based on encapsulation efficiency and survival of the entrapped probiotics after simulated gastrointestinal digestion; (ii) to characterize the optimized microbeads in terms of morphology, surface and internal structure, texture, thermal stability, and hydration-related properties; (iii) to assess the antioxidant activity and digestive enzyme inhibitory potential of the bioaccessible fractions released after in vitro digestion; and (iv) to examine the effects of the undigested fractions on strain-specific colonic fermentation responses, including pH and gas changes, SCFAs production, gut microbial community structure, predicted microbial functions, and fecal metabolomic profiles using 16S rRNA gene sequencing and untargeted LC-QTOF-MS analysis. Through this integrated approach, the study sought to evaluate Alg:PSE microbeads not only as protective probiotic carriers, but also as a synbiotic platform capable of modulating gut microbiota and fermentation metabolism. *Lactocaseibacillus rhamnosus* and *Bifidobacterium longum* subsp. *longum* were selected as

representative, widely studied probiotics of the *Lactobacillus* and *Bifidobacterium* groups with well-documented gastrointestinal health benefits; moreover, preliminary screening (Section 2.4) confirmed that both strains are acid-sensitive in their free form, making them suitable models for evaluating the protective performance of the encapsulation system.

2. Materials and methods

2.1. Materials and reagents

Whole date seeds (50 kg) were obtained from Al-Foah Date Processing Industry (Al Ain, Abu Dhabi, United Arab Emirates). The seeds were milled using a Retsch ZM200 DR 100 mill (Haan, Germany), defatted by two successive hexane extractions, and stored for subsequent analysis. Sodium alginate, calcium chloride, and other analytical-grade chemicals were purchased from Sigma-Aldrich Inc. (St. Louis, Missouri, USA), unless otherwise stated. Probiotic strains were obtained from Leibniz-Institut DSMZ-Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (Braunschweig, Germany).

2.2. Isolation of date seed polysaccharide

The ultrasound-assisted purified date seed polysaccharide fraction (UPS; Mw 3290 kDa) was prepared from defatted date seed powder according to the extraction procedure previously reported by Subhash, et al. [16]. Briefly, date seed polysaccharides were extracted using a choline chloride/ethylene glycol-based deep eutectic solvent system assisted by ultrasound, followed by protein removal using 8% trichloroacetic acid, ethanol precipitation, dialysis against water using a 20 kDa molecular-weight cut-off membrane, freeze-drying, and storage at -20°C until use. The purified UPS fraction was previously shown to be free from proteinaceous impurities and low-molecular-weight substances and was comprehensively characterized as a high-molecular-weight heteropolysaccharide. UPS had an average molecular weight of approximately 3290 kDa, as determined by gel permeation chromatography, and was mainly composed of galactose, mannose, fructose, glucose, and galacturonic acid at 66.1, 13.3, 9.9, 5.4, and 5.1%, respectively, indicating a galactose-rich acidic heteropolysaccharide structure. FTIR analysis showed characteristic polysaccharide bands, including broad O—H stretching around 3277 cm^{-1} , C—H stretching around 2997 cm^{-1} , uronic-acid/carboxyl-related bands around 1618, 1441, 1415, and 1382 cm^{-1} , and C—O/glycosidic vibrations around 1022 cm^{-1} . UPS also showed a particle size of 3303.6 nm, a strongly negative zeta potential, irregular porous morphology by SEM, a water solubility index of $82.9 \pm 0.7\%$, water-holding capacity of $2.9 \pm 0.1\text{ g/g}$, and oil-holding capacity of $6.4 \pm 0.3\text{ g/g}$. Rheological characterization indicated elastic-dominant behavior, with storage modulus (G') higher than loss modulus (G''), together with thixotropic tendencies. Based on these macromolecular and functional properties, UPS was used to prepare a 2.0 wt% polysaccharide solution for alginate reinforcement and pickering emulsion stabilization. The molecular weight, monosaccharide composition, and spectroscopic and functional properties reported above were determined in our previous study [16] and are summarized here only to define the material used; they were not re-measured in the present work. The high apparent molecular weight ($\sim 3290\text{ kDa}$) is consistent with a high-molecular-weight acidic heteropolysaccharide rather than protein contamination, as protein was removed by repeated 8% trichloroacetic acid treatment and the purified fraction eluted as a single principal peak in gel permeation chromatography with no protein-associated signal. The original GPC chromatogram and the full physicochemical characterization—including confirmation, by the phenol–sulfuric acid assay, that the purified fraction was free from proteins and low-molecular-weight impurities—were reported in our previous study [16], supporting this molecular-weight assignment.

2.3. Preparation of UPS-stabilized emulsion

An oil-in-water emulsion stabilized by UPS was prepared according to Li, et al. [10], with minor modification. A 2.0 wt% UPS solution was used as the aqueous emulsifier phase and mixed with corn oil at a weight ratio of 92:8. The mixture was homogenized using a T25 digital ULTRA-TURRAX at 22,000 rpm for 4 min to obtain a UPS-stabilized pickering emulsion, hereafter designated as PSE. The resulting emulsion was referred to as the UPS-stabilized emulsion (PSE). Therefore, in this study, PSE refers to the emulsion form of UPS, whereas UPS + alginate refers to direct mixing of polysaccharide with alginate without forming an emulsion. Because UPS is not fully molecularly dissolved but exists partly as hydrated colloidal particles (hydrodynamic size $\approx 3.3 \mu\text{m}$, irregular porous morphology, and elastic-dominant rheology; [16]), it can adsorb at the oil–water interface and act as a particulate stabilizer; on this basis the resulting system is referred to as a UPS-stabilized pickering emulsion. Direct interfacial characterization (droplet microscopy showing particle adsorption, three-phase contact angle, and interfacial tension measurements) would further confirm the pickering mechanism and is noted among the study limitations.

2.4. Preparation of probiotic suspensions

Lactocaseibacillus rhamnosus and *Bifidobacterium longum* subsp. *longum* DSMZ 20097 free cells were examined under pH 3.0 in acidified MRS at 37 °C for 3 h. For each strain, 100 μL of activated culture was transferred into 10 mL fresh MRS broth adjusted to pH 3.0 using HCl and incubated at 37 °C for 2 h. Samples were then serially diluted (10^{-1} – 10^{-7}) in sterile saline, plated on MRS agar, and incubated at 37 °C for 24–48 h. This procedure was conducted to ensure that both bacteria were acid sensitive. Afterward, *L. rhamnosus* and *B. longum* were propagated separately in MRS broth at 37 °C for 24 h, centrifuged at 4000 rpm for 15 min using a MiPC 12 centrifuge (MILAB, Dubai, UAE), and washed with sterile 0.1 M phosphate-buffered saline (PBS, pH 7.4). The resulting pellets were resuspended in sterile PBS to obtain a final concentration of approximately $12 \log_{10}$ CFU/mL.

2.5. Preparation of probiotic-loaded microbeads

Probiotic microbeads were prepared by ionic gelation according to Li et al. (2023), with minor modifications. A 2.0% (w/v) sodium alginate solution was used as the main encapsulating matrix. For each probiotic strain, five different formulations were prepared using alginate either alone or in combination with UPS or PSE.

For *L. rhamnosus*, the formulations were:

CL: alginate alone (control)
 APL: alginate + UPS (without emulsion)
 AET2: alginate + PSE at 1:1 (v/v)
 AET3: alginate + PSE at 1:2 (v/v)
 AET4: alginate + PSE at 1:3 (v/v)

For *B. longum*, the formulations were:

CB: alginate alone (control)
 APB: alginate + UPS (without emulsion)
 AET7: alginate + PSE at 1:1 (v/v)
 AET8: alginate + PSE at 1:2 (v/v)
 AET9: alginate + PSE at 1:3 (v/v)

For each formulation, 9.9 mL of the encapsulating mixture was combined with 0.1 mL of the corresponding probiotic suspension. The mixtures were vortexed for 15 min to ensure that the bacterial cells were distributed uniformly throughout the matrix. The resulting mixtures were then transferred into a 10 mL syringe fitted with a needle and extruded dropwise into a 5.0% (w/v) CaCl_2 solution. Continuous stirring

was applied during bead formation to help produce uniform microbeads. The beads were left in the CaCl_2 solution for 30 min to complete gelation and hardening. After that, the microbeads were collected by filtration and rinsed thoroughly with sterile distilled water to remove residual calcium ions from the surface. Freshly prepared microbeads were used immediately for probiotic viability determination and simulated gastrointestinal digestion studies. Additional aliquots of the beads were lyophilized and stored at -20°C for subsequent physicochemical characterization.

2.6. Selection of the optimal encapsulation systems

Encapsulation performance was evaluated based on viable counts before and after simulated gastrointestinal digestion. Freshly prepared microbeads (0.10 g) were dispersed in 1 mL of 0.1 M PBS (pH 7.4) and homogenized for 15 min with a sterile glass rod to promote bead disintegration through calcium chelation, according to Al-Hamayda, et al. [18]. Appropriate dilutions were plated on agar media and incubated in a 5% CO_2 incubator (Binder GmbH, Tuttlingen, Germany) at 37 °C for 24–48 h. Colonies were enumerated using a Scan 1200 HD automatic colony counter (Interscience, Paris, France), and results were expressed as $\log_{10}$ CFU/mL.

Gastrointestinal survival was assessed using the INFOGEST 2.0 static in vitro digestion model [18]. Briefly, 1 g of fresh microbeads was subjected to simulated oral, gastric, and intestinal digestion. At the end of digestion, surviving cells were enumerated as described above. Probiotic viability was calculated as:

$$\text{Survival rate (\%)} = 10^{(\log N_{\text{dig}} - \log N_{\text{enc}})} \times 100 \quad (1)$$

where N_{dig} is the number of viable cells after digestion and N_{enc} is the number of viable cells in freshly prepared encapsulated beads.

2.7. Structural and physicochemical characterization of selected microbeads

Based on encapsulation efficiency and digestive survival, AET3 and AET8 were selected as the optimal formulations for *L. rhamnosus* and *B. longum* subsp. *longum*, respectively, and compared with their corresponding alginate controls (CL and CB). Fresh and air-dried bead size was determined using an Olympus EP50 stereo microscope (Olympus Corporation, Tokyo, Japan) at $4.5\times$ magnification as described by Al-Hamayda, et al. [18]. Diameter, perimeter, and projected area were measured using Olympus EPview software.

Surface morphology and microstructure of lyophilized beads were examined by scanning electron microscopy (JEOL JSM-6010PLUS-LA, Akishima, Tokyo, Japan) after sputter coating. Images were acquired at $300\times$ to $4000\times$ magnification at an accelerating voltage of 20 kV. Mechanical properties were determined using a Brookfield CT3 texture analyzer (AMETEK Brookfield, Middleboro, USA) in compression mode with a target distance of 1.0 mm, trigger load of 6.8 g, test speed of 0.5 mm/s, two compression cycles, and a 4500 g load cell. Hardness, elasticity, cohesiveness, and resilience were calculated from 10 beads per treatment.

Molecular interactions between UPS and alginate were evaluated by ATR-FTIR using a Diamond/ZnSe crystal plate (PerkinElmer, Shelton, USA). Freeze-dried samples were scanned from 4000 to 400 cm^{-1} at 4 cm^{-1} resolution with 32 scans [10]. Crystalline characteristics were assessed by X-ray diffraction (Shimadzu-6100, Kyoto, Japan) using $\text{Cu-K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$) over a 2θ range of 5 – 100° . Thermal stability was assessed by thermogravimetric analysis and differential scanning calorimetry following [18]. TGA was conducted from 25 to 500°C at $10^\circ\text{C}/\text{min}$ under nitrogen ($30 \text{ mL}/\text{min}$), while DSC measurements were recorded from -45 to 250°C at $10^\circ\text{C}/\text{min}$. Thermogravimetric analysis and differential scanning calorimetry were performed using a TGA-2-star system (Mettler Toledo, Switzerland) and a TA Instruments (New

Castle, DE, USA), respectively.

2.8. Moisture content and swelling behavior

Moisture content of fresh microbeads was determined according to AOAC method 925.10 [19]. Approximately 5 g of each sample was dried at 105 °C to constant weight. Moisture content was calculated as:

$$\text{Moisture (\%)} = \frac{\text{Mass of moisture loss}}{\text{Original sample mass}} \times 100 \quad (2)$$

Swelling behavior of lyophilized microbeads was assessed according to Phùng, et al. [20]. Approximately 25 mg of beads was immersed in 10 mL sterile ultrapure water and maintained at room temperature for 6 h. Samples were weighed hourly, and swelling ratio was calculated as:

$$\text{Swelling ratio} = \frac{W_t}{W_o} \quad (3)$$

where W_t is the weight of swollen beads at a given time and W_i is the initial dry weight.

2.9. Simulated gastrointestinal digestion and bioactivity assays

Fresh microbeads were subjected to the standardized INFOGEST 2.0 static digestion model [18]. Briefly, 1 g of sample was gently crushed to simulate oral mastication and mixed with simulated salivary fluid containing amylase (75 U/mL) for 2 min. Gastric digestion was then carried out in simulated gastric fluid (pH 3.0) containing pepsin (2000 U/mL) and gastric lipase (60 U/mL) for 120 min. This was followed by intestinal digestion in simulated intestinal fluid (pH 7.0) containing pancreatin (100 U/mL) and bile salts (10 mmol/L) for a further 120 min.

After digestion, samples were centrifuged to separate the bio-accessible and non-digested fractions. The supernatants were used for antioxidant and enzyme inhibition analyses, whereas the residual fractions were stored at −20 °C for fecal fermentation experiments. Antioxidant capacity (FRAP, TAC, and RP), radical scavenging activities (DPPH, ABTS, SOD, and SAS), and enzyme inhibition assays (α -amylase, α -glucosidase, and ACE inhibition) were performed according to Subhash, et al. [17]. Radical scavenging activity was expressed as percentage inhibition, FRAP, RP, and TAC were expressed as μ g/mL ascorbic acid equivalents, and enzyme inhibition was calculated relative to the corresponding controls.

2.10. In vitro fecal fermentation and microbiota-related analyses

2.10.1. Fermentation design and monitoring

The optimal formulations AET3 and AET8 were selected for in vitro fecal fermentation. The residual undigested fractions obtained after simulated digestion were used as fermentation substrates following [16], with minor modification. A galacto-oligosaccharide preparation at the same concentration as UPS served as the positive control (AEP), whereas basal medium without an added carbon source served as the negative control (AEN). Fresh fecal samples were collected from healthy adult volunteers ($n = 6$; 3 men and 3 women, aged 30–40 years) with no antibiotic use during the preceding three months, under approval from the UAEU Human Research Ethics Committee (No. ERH_2024_5441). A pooled 20% (w/v) fecal slurry was prepared in 0.01 M phosphate buffer (pH 7.0). Each anaerobic fermentation vessel had a total volume of 25 mL and contained 10% fecal slurry, 86% sterilized basal medium, and 4% of the test sample, GOS-P, or control substrate. Fermentations were incubated anaerobically at 37 °C, and pH and gas production were recorded at 0, 6, 12, and 24 h.

2.10.2. SCFA determination

At each fermentation time point, aliquots from AEN, AEP, AET3, and AET8 were centrifuged at 15,000 $\times g$ for 20 min, and the supernatants

were filtered through 0.45 μ m membranes. SCFAs were quantified according to Subhash, et al. [17] using a Shimadzu HPLC system equipped with an SPD-M20A photodiode array detector and a Shodex C18M 4E column (250 $\times$ 4.6 mm, 5 μ m). The mobile phase consisted of 10 mM monopotassium phosphate (pH 2.4) and acetonitrile (80:20, v/v), with isocratic elution at 1.5 mL/min. The column temperature was maintained at 30 °C, injection volume was 20 μ L, and detection was performed at 210 nm. Acetic, propionic, and butyric acids were quantified using external calibration.

2.10.3. Microbial community profiling

Changes in microbial composition in AEN, AEP, AET3, and AET8 during fermentation were investigated using 16S rRNA gene amplicon sequencing. Genomic DNA was extracted using a Tiangen Genomic DNA Kit (Beijing, China). The V3–V4 region of the bacterial 16S rRNA gene was amplified, and sequencing was performed at BGI (Hong Kong, China). PCR products were purified using Agencourt AMPure XP beads, and library size and concentration were checked using an Agilent 2100 Bioanalyzer.

Raw reads were quality filtered with iTools Fqtools fqcheck (v0.25), and paired-end reads with overlapping regions were merged using FLASH (v1.2.11). Tags were clustered into operational taxonomic units (OTUs) at 97% similarity using USEARCH (v7.0.1090), and chimeric sequences were removed with UCHIME (v4.2.40). Representative OTU sequences were assigned taxonomically using the RDP classifier (v2.2) at a 60% confidence threshold. Shared and unique OTUs were visualized by Venn diagrams in R (v3.1.1). Alpha diversity was evaluated using standard richness and diversity indices, while beta diversity was assessed using community dissimilarity metrics. Multivariate analyses, including PCA, were performed using the ade package in R, and differential taxa were identified by LEfSe on the Galaxy platform. Functional prediction of microbial communities, including KEGG, COG, and MetaCyc pathway annotation, was performed using PICRUSt2 (v2.3.0-b). Correlation analyses and microbial interaction visualizations were generated in R and Cytoscape based on Spearman correlation coefficients.

2.11. Untargeted LC–MS metabolomic analysis

2.11.1. Sample preparation and LC–MS conditions

After 24 h fecal fermentation, aliquots from AEN, AEP, AET3, and AET8 were concentrated in a CentriVap concentrator at 35 °C for 16 h. Concentrated samples (200 μ L) were extracted with 800 μ L LC–MS-grade acetonitrile, filtered through 0.22 μ m membranes, and subjected to untargeted metabolomic profiling.

Chromatographic analysis was performed using a Waters ACQUITY UPLC system equipped with a Sample Manager FTN and Binary Solvent Manager, coupled to a Vion IMS QToF mass spectrometer (Waters, Milford, MA, USA). Separation was achieved on an ACQUITY UPLC BEH C18 column (2.1 $\times$ 50 mm, 1.7 μ m). The column temperature was maintained at 40 °C and the sample temperature at 10 °C. The mobile phases consisted of solvent A (water containing 0.1% formic acid) and solvent B (acetonitrile containing 0.1% formic acid). Elution was performed at a flow rate of 0.45 mL/min using the following gradient: 0.00–2.00 min, 98% A/2% B; 2.50 min, 93% A/7% B; 9.00 min, 50% A/50% B; 13.00 min, 15% A/85% B; 19.50 min, 5% A/95% B; and re-equilibration to 98% A/2% B at 22.00 min.

Mass spectrometric detection was conducted in positive electrospray ionization mode using MS^E acquisition in sensitivity mode with soft transmission. Data were acquired over an m/z range of 50–1000 with a scan time of 0.150 s. The low collision energy was set at 6 V and the high-energy collision ramp was set from 20 to 45 V. Source parameters were as follows: source temperature, 120 °C; desolvation temperature, 450 °C; cone gas, 50 L/h; desolvation gas, 800 L/h; capillary voltage, 0.80 kV; sample cone voltage, 20 V; and source offset voltage, 80 V. Automatic lock-mass correction was enabled throughout data

acquisition, with Leucine Enkephalin used as the lock-mass standard for real-time mass correction. Prior to sample analysis, the instrument was calibrated using the Waters Vion Test Mix, a nine-component reference standard containing acetaminophen, caffeine, sulfaguanidine, sulfadimethoxine, Val-Tyr-Val, verapamil, terfenadine, Leucine Enkephalin, and reserpine.

To monitor analytical stability and reduce signal drift, pooled quality control (QC) samples were included during the analytical sequence and analyzed after every 10 samples. In addition, the acquisition sequence defined QC, blank, reagent blank, standard, and reference sample types to support data-quality assessment. Raw data were processed using Progenesis QI (version 2.2) for peak alignment, peak extraction, normalization, deconvolution, and compound annotation. Peak alignment was performed using the most suitable QC sample as the reference.

Peak processing was carried out using automatic integration and threshold settings with mean smoothing (half width = 2; iterations = 2). Low- and high-energy intensity thresholds were set at 150 and 50 counts, respectively, and lock-mass correction was applied during feature extraction.

2.11.2. Data processing and metabolite annotation

Putative metabolite annotation was performed by matching accurate mass values and corresponding MS^E fragmentation patterns against the HMDB and METLIN databases. A mass deviation threshold of 10 ppm was applied during database searching. Multivariate analyses, including principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA), were performed together with variable importance in projection (VIP) scoring and univariate statistical

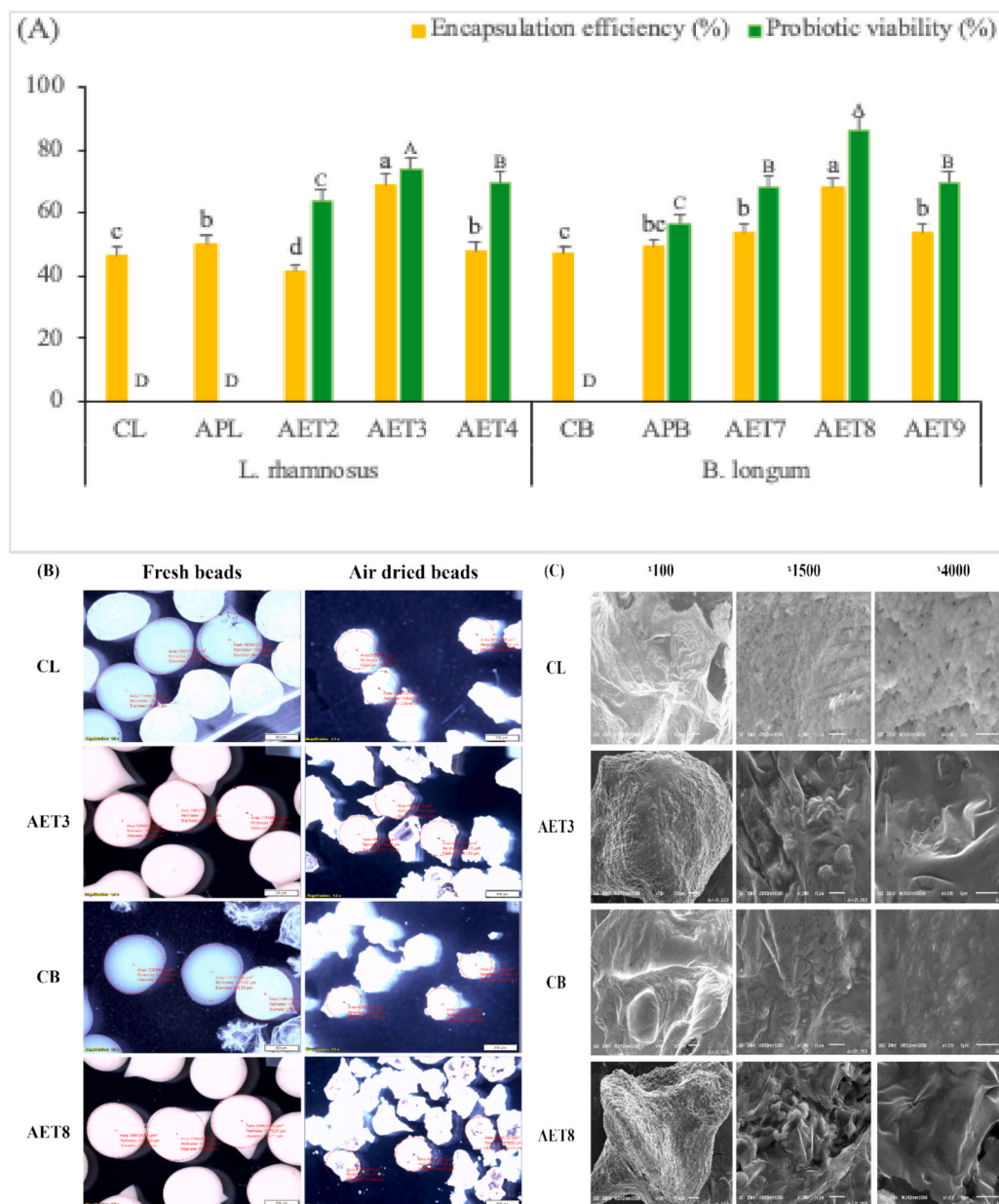


Fig. 1. Encapsulation performance and morphology of probiotic-loaded microbeads. (A) Encapsulation efficiency (%) and probiotic viability (%) after simulated gastrointestinal digestion of *L. rhamnosus* microbeads (CL, APL, AET2, AET3, and AET4) and *B. longum* subsp. *longum* microbeads (CB, APB, AET7, AET8, and AET9). (B) Representative stereomicroscopy images of selected formulations before and after air-drying. From top to bottom: CL, AET3, CB, and AET8; left column, fresh beads; right column, air-dried beads. (C) Scanning electron micrographs of lyophilized CL, AET3, CB, and AET8. For each formulation, images are shown at increasing magnification from left to right. Different lowercase letters indicate significant differences among treatments within the same parameter at $P < 0.05$. Data in panel A are presented as mean $\pm$ SD ($n = 10$). The mean comparison between treatments was performed for each parameter at the same bacteria.

testing. Metabolite features with $VIP > 1$ and $P < 0.05$ were considered differential. Because metabolite annotation was based on database matching rather than confirmation with authentic chemical standards, the reported identities should be considered putative. No additional multiple-testing correction was applied; therefore, metabolite-level differences were interpreted together with VIP scores, clustering behavior, and pathway-enrichment patterns. Data processing was performed using MetaboAnalyst 6.0 (<https://www.metaboanalyst.ca/>), MicrobiomeAnalyst (<https://www.microbiomeanalyst.ca/>), and OmicShare (<https://www.omicshare.com/tools/>). In accordance with the Metabolomics Standards Initiative (MSI) reporting guidelines, annotations based on accurate mass and MSE matching without authentic chemical standards correspond to putative (MSI level 2–3) identifications and are reported as putative throughout. Accordingly, the discriminant features were interpreted together with their VIP scores, clustering behavior, and pathway-enrichment patterns rather than as definitive structural identifications.

2.12. Statistical analysis

Experiments were performed in triplicate, with results reported as mean $\pm$ SD. Analyses used RStudio (R version 2026.01.0 + 392). Datasets were checked for normality and equal variance before analysis. Treatment differences were assessed with one-way or two-way ANOVA based on design, followed by Tukey's post hoc test for multiple comparisons if effects were significant. Significance was set at $P < 0.05$.

3. Results and discussion

3.1. Probiotic viability and encapsulation efficiency

Fig. 1A revealed that the efficiency of encapsulation and probiotic cells' survivability in capsules after simulated digestion were highly dependent on the type of matrix. In terms of *L. rhamnosus*, alginate (CL) and alginate-UPS (APL) matrices did not show differences in their encapsulation efficiency (46.8–49.8%). Neither CL nor APL matrices contained surviving cells after digestion. Incorporation of the UPS-stabilized emulsion significantly boosted performance of microbeads. Among the emulsion-containing matrices, the highest encapsulation efficiency ($\sim 68.9\%$) and post-digestion survival (74.0%) were obtained using AET3 (Alg:PSE2). Only 0.2 log reduction was observed for cells treated with INFOGEST in this matrix. AET2 and AET4 had beneficial influence on bacterial survivability but to a lesser degree, with 64.2% and 69.7% survival rates, correspondingly.

A similar tendency was observed for *B. longum* subsp. *longum*. UPS had less positive effect on this bacterium compared to *L. rhamnosus*. While the alginate control had low encapsulation efficiency ($\sim 47\%$) and lack of viable cells after digestion, other matrices, including all emulsion-containing ones, had some level of survival. Highest survivability was observed for AET8 (Alg:PSE2), where post-digestion percentage reached 86.3%, which was more efficient compared to AET7 with $68.5 \pm 7.71\%$ and AET9 with 69.5%. Therefore, for both types of bacteria, the 1:2 Alginate:PSE ratio appeared to be optimal for achieving the most beneficial combination of cell encapsulation and protection from the harshness of the digestive tract.

These results imply that the presence of the pickering emulsion is more than just a contribution to increased solid concentration; it can also affect the properties of alginate gel structure and create an additional hydrogen ion, digestive enzyme, and bile salt diffusion barrier. The mentioned effect correlates with other previous studies reporting decreased porosity of alginate microgels through filling with polymeric and colloidal substances (Li et al., 2023; Riaz et al., 2019; Zhu et al., 2025). Similar observations were made with regard to fillers added into alginate-based probiotic carriers (Li et al., 2023; Zhang et al., 2023; Zhang et al., 2025). Thus, according to Li et al. (2023), pure alginate-based matrix could not provide enough protection to *Lactobacillus*

casei from digestive processes, whereas polymeric and emulsion fillers helped to increase cell survivability. Also, (Zhang et al., 2021) has shown increased retention of *Bifidobacterium pseudocatenulatum* in alginate microgels containing both $CaCO_3$ and lipid nanoemulsion droplets, while (Riaz et al., 2019) found that coating alginate beads improved the gastrointestinal survival of *Bifidobacterium bifidum*. Our data also indicate that alginate alone was insufficient under the present digestion conditions. This agrees with (Guo et al., 2024), who observed no detectable survivors in alginate control beads after intestinal digestion, and with (Ismail et al., 2025), who noted that alginate-based systems can show limited encapsulation performance unless structurally reinforced. The particularly strong effect of AET3 and AET8 therefore supports the use of UPS-stabilized emulsions as functional fillers that increase matrix compactness and preserve viable cells during gastrointestinal transit. The encapsulation efficiencies obtained here (~ 47 – 69%) lie within the moderate range commonly reported for alginate-based probiotic beads produced by ionic gelation/extrusion, in which the relatively large bead size, the porosity of pure alginate, and cell losses during gelation and washing typically limit entrapment unless the matrix is reinforced [10]. Consistent with this, the lowest efficiencies corresponded to the unreinforced alginate and alginate-UPS systems, whereas incorporation of the PSE raised the value to $\sim 68.9\%$ (AET3), indicating that the emulsion filler improves cell retention primarily by densifying the matrix.

3.2. Structural and physicochemical characterization of microcapsules

Optimized formulations (AET3 and AET8) were compared with their alginate-based counterparts (CL and CB). In the fresh state, beads of all formulations maintained their spherical shape; however, upon air drying, there was a decrease in the area, perimeter, and diameter parameters in all treatments, indicative of contraction of the polymer matrix following water release (see Table S1 and Fig. 1B). Bead areas in fresh condition varied from 0.107 to 0.117 mm², while in air-dried condition, they decreased to the range 0.038–0.047 mm². Nonetheless, despite the decrease in size, the UPS-emulsion beads appeared more compact after drying when compared with the alginate-based counterparts, implying the creation of a denser internal structure due to emulsion incorporation.

This conclusion was supported by SEM images showing the difference in surface architecture between control (Fig. 1C) and optimized beads (AET3 and AET8). Alginate-based control beads featured rough, folded, and relatively porous surfaces, whereas the optimized beads contained smoother, denser, and more homogenous matrices with less apparent porosity and fissures. These differences are indicative of the space-filling role played by the emulsions in the alginate network along with the increased number of interfacial interactions between UPS and the polymeric matrix [10,21]. Similar bead morphologies containing alginate-based composite matrices with polysaccharides or emulsion-filled cavities have already been described in the literature [10,21].

Moreover, FTIR spectroscopy revealed matrix reinforcement in the optimized formulations (Fig. 2A). Characteristic peaks in all beads corresponded to the presence of C–H aliphatic stretching at 2929 and 2856 cm⁻¹, C=O stretching of esters or carbonyls at 1746 cm⁻¹, C–O–C stretching of polysaccharides at 1163 cm⁻¹, and alginate carboxylate group at 1627 cm⁻¹. In this respect, the increased intensity of some hydrophobic and carbonyl-related bands in AET3 and AET8, especially in AET3, is indicative of an increased amount of non-covalent interactions, as well as the incorporation of UPS-emulsions in the alginate matrix. These conclusions are in agreement with the data obtained using SEM.

Finally, thermal analyses have also allowed distinguishing the optimized formulations from controls (Fig. 2B,C). DSC thermograms have shown that CL and CB demonstrated relatively sharp peaks between ~ 80 – 160 °C, corresponding to water loss and polymer relaxation processes. Meanwhile, in the case of AET3 and AET8, a wide endothermic transition could be observed. The TGA thermograms demonstrated a

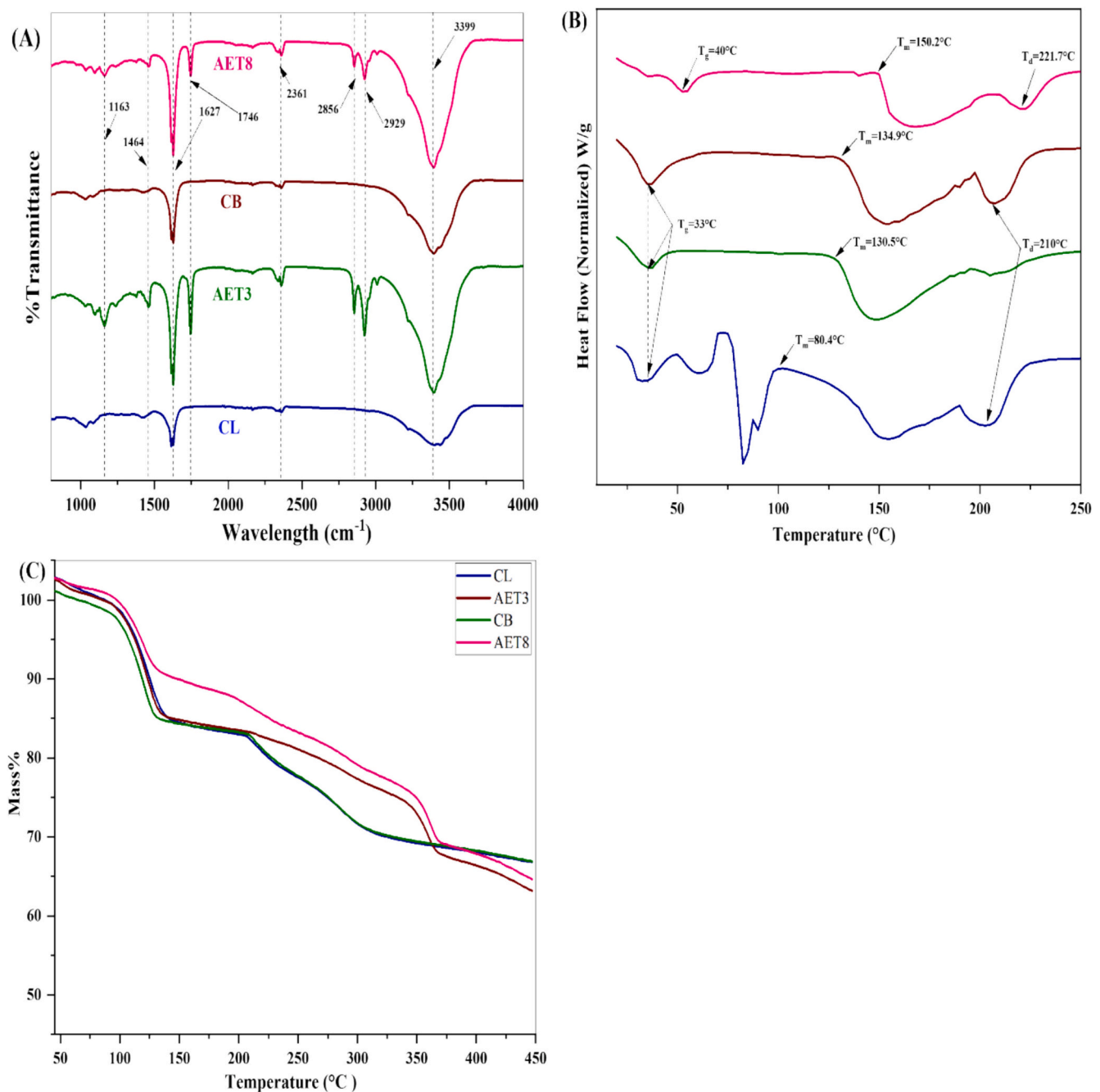


Fig. 2. Structural and thermal characterization of selected microbeads. (A) FTIR spectra of CL, AET3, CB, and AET8 showing the characteristic functional groups and molecular interactions within alginate- and UPS-emulsion-based matrices. (B) Differential scanning calorimetry (DSC) thermograms of CL, AET3, CB, and AET8 showing thermal transitions and differences in heat-induced structural changes among formulations. (C) Thermogravimetric analysis (TGA) curves of CL, AET3, CB, and AET8 showing thermal degradation profiles and relative thermal stability of the microbeads.

multi-stage degradation pattern in all samples, with initial weight loss corresponding to water release. However, the major part of the thermal decomposition occurred sooner in control beads and later in optimized beads. The latter appeared to demonstrate higher stability at temperatures >350 °C. This can be explained by more stable and dense structure of composite matrix, caused by stronger polymer-polysaccharide interactions [22,23]. The greater thermal stability of AET3 and AET8 can be explained by the denser, less porous composite network formed when the PSE fills the alginate matrix: extensive hydrogen bonding between UPS and alginate, together with the lipid phase of the emulsion, restricts polymer-chain mobility and slows volatilization and degradation, shifting the major decomposition step to higher temperatures (>350 °C)

relative to the unreinforced controls.

Additionally, texture profile analysis corroborated these structural differences (Table 1). In terms of hardness, AET3 appeared to be the strongest (231.5 ± 11.81 g), followed by AET8 (219.3 ± 11.48 g). Controls showed significantly lower hardness, being 156.8 ± 31.66 g and 152.86 ± 32.21 g for CL and CB, respectively. In contrast, the controls appeared more elastic, cohesive, and resilient than the UPS-emulsion beads. As follows from the results, the addition of PSE to the matrix has led to the increased mechanical strength but reduced elastic deformability. For probiotic carriers, this property is advantageous because the more rigid matrix is more resistant to deformation and fluid penetration both in the manufacturing process and in the conditions of

Table 1

Texture profile parameters of microbead formulations.

Treatment ¹	Hardness (g)	Elasticity (mm)	Cohesiveness	Resilience
CL	156.8 ± 31.66 ^c	0.276 ± 0.13 ^b	0.211 ± 0.32 ^b	0.070 ± 0.01 ^a
AET3	231.5 ± 11.81 ^a	0.073 ± 0.48 ^c	0.010 ± 0.12 ^c	0.008 ± 0.01 ^b
CB	152.86 ± 32.21 ^c	0.546 ± 0.11 ^a	0.520 ± 0.67 ^a	0.060 ± 0.11 ^a
AET8	219.3 ± 11.48 ^b	0.086 ± 0.12 ^c	0.023 ± 0.09 ^c	0.006 ± 0.00 ^b

(a-c) Mean values ± Standard deviations that do not share a letter in the same column are significantly different ($P < 0.05$).¹ CL and CB: control beads of alginate alone encapsulating *L. rhamnosus* and *B. longum* subsp. longum, respectively; AET3 and AET8: alginate:PSE (1:2, v/v; Alg:PSE2) microbeads encapsulating *L. rhamnosus* and *B. longum* subsp. longum, respectively. PSE, UPS-stabilized Pickering emulsion; UPS, ultrasound-extracted date seed polysaccharide.

digestion. The greater mechanical strength of AET3 and AET8 is therefore attributed to the space-filling and matrix-densifying effect of the incorporated PSE rather than to specific biopolymer–cell-wall interactions, which were not examined in the present study. It should also be noted that texture was measured on fully hydrated beads (moisture 90–97%); the reported parameters therefore reflect the mechanical behavior of the hydrated delivery system, and absolute values should be interpreted in that context.

3.3. Swelling indices and moisture retention properties

The differences in hydration between the controls and PSE-containing microbeads are summarized in Table S2. Fresh alginate beads contained higher amounts of moisture: CL had moisture content of $97.0 \pm 0.36\%$, and CB – $95.5 \pm 0.19\%$. In contrast, the values for AET3 and AET8 were $90.7 \pm 0.15\%$ and $91.6 \pm 0.19\%$, respectively. The same trends were observed for swelling: CL and CB had swelling ratios of 3.1 ± 0.01 and 2.8 ± 0.01 , respectively, whereas those of AET3 and AET8 were significantly lower at 2.0 ± 0.01 and 1.9 ± 0.06 .

Reduced uptake and swelling imply lower porosity of the Alg:PSE matrix, which hindered the hydration of the gel. Such observations agree well with the conclusions drawn from SEM, FTIR, and thermal characterization, all indicating formation of a more dense and hydrophobic gel matrix. Importantly, such matrix characteristics promote slower penetration of digestive juices and increased retention of entrapped bacteria in probiotic delivery systems [24,25]. It can also be assumed based on prior experience with the use of polysaccharide emulsions or nanoemulsions in hydrogel systems when the dispersed phase serves as a space filler and barrier to hydration of the matrix [26–28]. The results obtained provide a partial explanation of the superior post-digestion viability of AET3 and AET8.

3.4. Bioactivities after in vitro digestion

As expected, the bioaccessible fractions obtained after in vitro digestion demonstrated higher antioxidant activities and enzymatic inhibitions in UPS-emulsion-based formulations as compared to the alginate counterparts (Fig. S1A–C). Regarding the radical scavenging capacity, the most efficient in terms of DPPH scavenging ability was AET3 ($63.0 \pm 1.34\%$), while the highest ABTS ($57.9 \pm 0.34\%$) and superoxide anion ($31.9 \pm 0.41\%$) scavenging capacities belonged to AET8. Moreover, both formulations surpassed the controls in terms of SOD activity, FRAP, and total antioxidant capacity. For instance, FRAP was equal to 64.4 ± 1.26 mg/L in case of AET8 and 56.5 ± 1.10 mg/L for AET3, which were much higher than in case of CL and CB. Likewise, the enzyme inhibition activities followed the similar pattern, with the maximum α -amylase (AET8– $48.8 \pm 1.38\%$) and α -glucosidase (AET3– $43.7 \pm 4.87\%$) inhibitors found in UPS-based microbeads, while the activities of ACE inhibitory activities were almost doubled compared to the controls (Fig. S1C).

The obtained results correlate well with the double function of UPS as bioactive polysaccharide and the building block for microcapsules. In our previous study, we have shown the presence of high antioxidant and

inhibitory activity of date seed polysaccharides [16]. In this regard, the direct contribution of UPS to the bioactivity of the released fraction as well as a protective effect of denser microcapsules on the bioavailability of functional metabolites could be expected after in vitro digestion of microcapsules. In this sense, the results are consistent with our previous studies of polysaccharide and probiotic co-encapsulation, which led to increased bioavailability of antioxidants due to polysaccharide or polyphenol contribution or emulsion-based carrier utilization [24,25,29,30].

3.5. In vitro fecal fermentation and modulation of the gut microbiota

3.5.1. Parameters of fecal fermentation and SCFAs synthesis

The impact of AET3 and AET8 on fecal fermentation parameters has been investigated using the negative control (AEN) and the GOS-P positive control (AEP). All fermentation systems experienced pH drop over 24 h; however, the extent of the acidification depended on the substrate (Table 2). The AEP produced the lowest pH values at 6, 12, and 24 h (6.23–6.25); hence, this substrate induced the most sustained fermentation. In turn, AET3 and AET8 remained least acidic, ranging from 6.48 to 6.61, while AET8 was characterized by the weakest acidification. Different trends can be seen when gas production has been studied. There was no detected gas during the initial period; however, after 6 h, both AET3 and AET8 systems produced greater amounts of gases (2.98 mL and 3.38 mL, respectively) compared to AEP (2.20 mL) and AEN (2.08 mL). At 24 h, the maximal amount of gas was observed in the AEP (7.13 mL) followed by AET8 (6.08 mL), AET3 (5.85 mL), and AEN (5.30 mL). As follows, the use of microbeads caused acceleration of fermentation compared to GOS-P despite being less active at the final stage. It should be noted that the negative (AEN, carbon-source-free) and positive (AEP, GOS) controls follow the conventional design of in vitro fecal fermentation; bare alginate is poorly fermented by the colonic microbiota, and the prebiotic activity of the matrix polysaccharide (UPS) itself was independently established in our previous work, where UPS alone stimulated SCFA production and enriched *Bifidobacterium* and *Blautia* [16]. The contributions of the matrix and of the released probiotics can therefore be interpreted in light of these references.

SCFAs generation was dependent on time and type of substrate used (Table 2). It is worth mentioning that acetate was the most abundant acid, with AET3 and AET8 having the largest initial concentration (799.04 ± 7.61 $\mu\text{mol/L}$ and 725.40 ± 10.52 $\mu\text{mol/L}$, respectively). However, all systems showed a decrease in acetate content; at the same time, at 12 h and 24 h, AEP still possessed the largest amounts of this substance (431.15 ± 5.94 $\mu\text{mol/L}$ and 397.62 ± 0.19 $\mu\text{mol/L}$, respectively). Propionate also demonstrated a decrease in its content with time; nonetheless, at initial stages of the experiment, the highest concentration was found in AET3 (152.09 ± 3.52 $\mu\text{mol/L}$), and at 6 h, the highest level was found in AET8 (114.52 ± 0.16 $\mu\text{mol/L}$), while at 24 h, AEP possessed the highest remaining amount (43.95 ± 0.72 $\mu\text{mol/L}$). Butyrate showed the strongest effect of probiotics added to the system. Thus, at 6 h, AET3 and AET8 contained higher amounts of butyrate (10.02 ± 0.86 $\mu\text{mol/L}$ and 7.79 ± 0.44 $\mu\text{mol/L}$, respectively) compared to other substrates (AEP – 3.28 ± 0.05 $\mu\text{mol/L}$; AEN – 1.13 ± 0.34 $\mu\text{mol/L}$).

Table 2

Parameters of the in vitro fecal fermentation.

Time(h)	Samples	pH	Gas (mL)	AA (μmol/L)	PA (μmol/L)	BA (μmol/L)
0	AEN	6.71 ± 0.01 ^b	ND	691.57 ± 3.34 ^d	119.78 ± 2.24 ^d	24.84 ± 2.30 ^b
	AEP	6.77 ± 0.01 ^a	ND	699.71 ± 5.78 ^c	132.12 ± 1.70 ^b	24.36 ± 1.73 ^b
	AET3	6.71 ± 0.01 ^b	ND	799.04 ± 7.61 ^a	152.09 ± 3.52 ^a	27.88 ± 1.53 ^a
	AET8	6.70 ± 0.01 ^c	ND	725.40 ± 10.52 ^b	131.97 ± 4.36 ^c	24.67 ± 0.68 ^b
6	AEN	6.48 ± 0.01 ^c	2.08 ± 0.11 ^d	605.28 ± 0.24 ^a	95.46 ± 0.39 ^c	1.13 ± 0.34 ^d
	AEP	6.23 ± 0.01 ^d	2.20 ± 0.07 ^c	471.01 ± 0.24 ^c	91.31 ± 0.81 ^d	3.28 ± 0.05 ^c
	AET3	6.55 ± 0.01 ^b	2.98 ± 0.04 ^b	568.31 ± 0.68 ^b	114.52 ± 0.16 ^a	10.02 ± 0.86 ^a
	AET8	6.61 ± 0.07 ^a	3.38 ± 0.18 ^a	395.38 ± 20.25 ^d	101.11 ± 0.98 ^b	7.79 ± 0.44 ^b
12	AEN	6.48 ± 0.01 ^b	2.43 ± 0.11 ^d	392.99 ± 7.27 ^b	87.35 ± 1.52 ^c	2.67 ± 0.18 ^c
	AEP	6.24 ± 0.01 ^c	6.28 ± 0.04 ^a	431.15 ± 5.94 ^a	96.37 ± 0.27 ^a	4.94 ± 0.73 ^a
	AET3	6.53 ± 0.01 ^a	4.58 ± 0.11 ^b	376.64 ± 0.86 ^c	93.41 ± 1.27 ^b	3.27 ± 0.65 ^b
	AET8	6.48 ± 0.01 ^b	4.08 ± 0.11 ^c	372.90 ± 7.65 ^d	87.13 ± 3.27 ^c	3.61 ± 0.75 ^b
24	AEN	6.51 ± 0.01 ^b	5.30 ± 0.07 ^d	348.60 ± 4.33 ^b	38.65 ± 0.03 ^b	3.50 ± 0.33 ^c
	AEP	6.25 ± 0.01 ^c	7.13 ± 0.18 ^a	397.62 ± 0.19 ^a	43.95 ± 0.72 ^a	5.76 ± 1.99 ^a
	AET3	6.51 ± 0.01 ^b	5.85 ± 0.14 ^c	309.24 ± 4.29 ^d	38.40 ± 1.22 ^b	3.82 ± 0.35 ^c
	AET8	6.52 ± 0.01 ^a	6.08 ± 0.11 ^b	321.39 ± 6.95 ^c	38.95 ± 1.59 ^b	4.75 ± 0.23 ^b

(a-c) Mean values ± standard deviations with different superscripts in the same column are significantly different ($p < 0.05$). AEN represents negative control, AEP represents positive control, GOS-P, AET3 represents *Lactococcus rhamnosus* encapsulated microbeads, and AET8 represents *Bifidobacterium longum* subsp. *longum* encapsulated microbeads after 24 h of fermentation; AA: Acetic acid; PA: Propionic acid, and BA: Butyric acid.

L), and at 24 h, AEP produced the highest amount of this SCFA ($5.76 \pm 1.99 \mu\text{mol/L}$).

As one can see from the above-discussed findings, AET3 and AET8 cause induction of early fermentation of a specific nature, with propionate and butyrate formation, whereas GOS-P results in sustained fermentation. Such a discrepancy can be explained by the fact that the smaller drop in pH in the former two systems does not mean a lack of activity; instead, it indicates the creation of a more buffered fermentation environment. Since polysaccharides of dates were included into the alginate-based matrix in microbeads, their release might have been slower than for freely dissolved GOS-P; thus, the process of microbead digestion led to quick start of fermentation. Indeed, such substrate-

dependent changes have been shown for other encapsulated probiotic systems and polysaccharide-supplemented fermentations [31,32]. Considering the significance of butyrate for energy supply of colonocytes and of propionate for metabolic signaling, this finding can be considered crucial [7]. Lactate was not retained as a major end-product, which is expected because the lactate generated by lactic acid bacteria such as *L. rhamnosus* is a transient intermediate that is rapidly cross-fed and converted by the microbiota into butyrate and propionate; the pronounced early butyrate and propionate observed here are consistent with such lactate cross-feeding. Comparable in vitro fecal-fermentation studies of dietary polysaccharides have likewise reported substrate-driven SCFA generation with early enrichment of propionate and

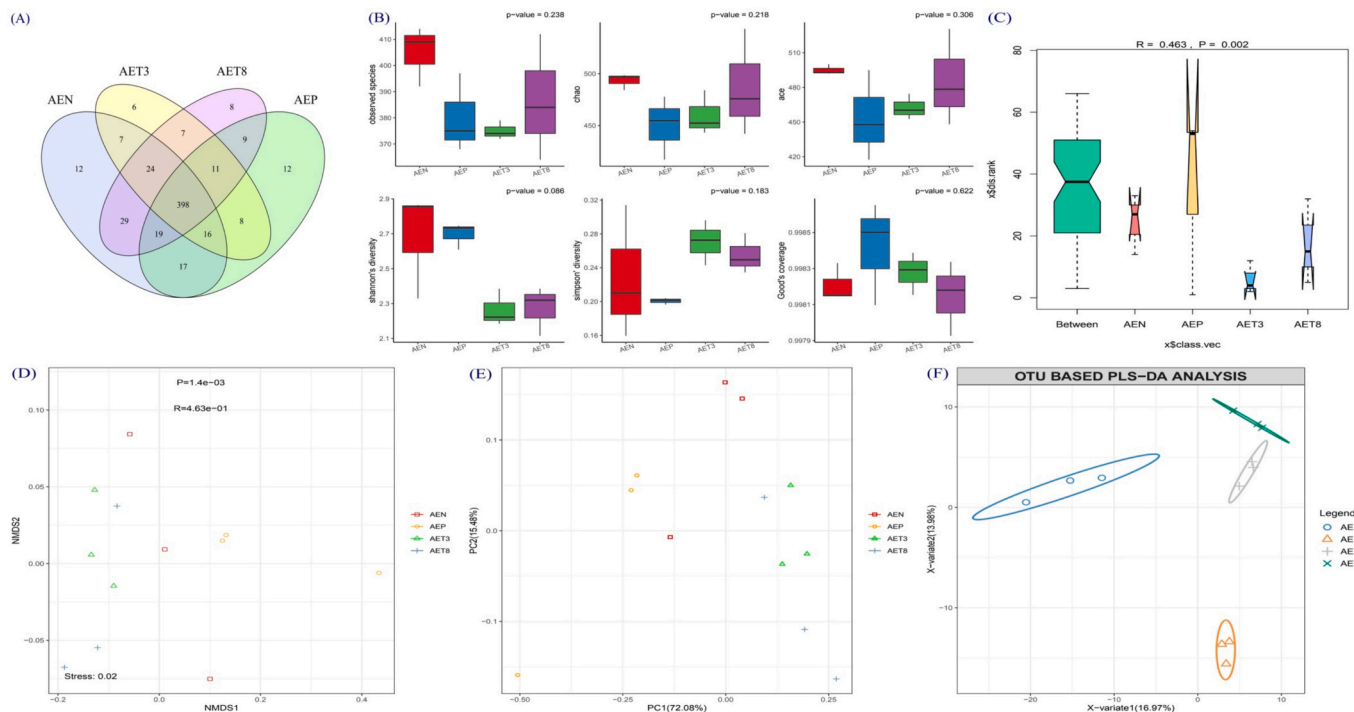


Fig. 3. Effects of the selected microbeads on gut microbial diversity after 24 h in vitro fecal fermentation. (A) Four-way Venn diagram showing shared and unique OTUs among AEN, AEP, AET3, and AET8. (B) Alpha-diversity boxplots showing observed species, Chao, ACE, Shannon, Simpson, and Good's coverage indices. (C) ANOSIM analysis showing between-group and within-group dissimilarity. (D) Non-metric multidimensional scaling (NMDS) plot showing differences in microbial community structure among treatments. (E) Taxon-level principal component analysis (PCA) plot. (F) OTU-based partial least squares discriminant analysis (PLS-DA) plot showing treatment-wise clustering. AEN, negative control; AEP, GOS-P positive control; AET3, *L. rhamnosus* encapsulated in alginate + PSE (1:2, v/v); AET8, *B. longum* subsp. *longum* encapsulated in alginate + PSE (1:2, v/v).

butyrate, supporting our findings [33,34].

3.5.2. Microbial diversity and community structure

The Venn diagram revealed considerable overlap of the core microbiota in different treatments with 398 OTUs present in all four tested groups – AEN, AEP, AET3, and AET8 – whereas the unique OTUs were limited (Fig. 3A). This suggests that the probiotics had no effect on alpha diversity/richness but influenced its structure.

Analysis of alpha diversity did not reveal any changes between groups (Fig. 3B). None of the metrics – observed species, Chao, ACE, Shannon, Simpson, or Good's coverage returned statistically significant differences between the groups, whereas the Good's coverage value approached 1.0, thus indicating adequate sequencing depth. This was confirmed with rarefaction plots (Fig. S2A–F). Even though the highest richness and diversity were measured in AEN, and the lowest – in AET3, the lack of significance in alpha diversity metrics implies that the main action of encapsulated probiotics was based on community composition alteration. Similar observations were made in other studies using probiotics encapsulated in microbeads, when changes in microbial community structure were achieved regardless of significant alterations in the diversity [14,35]. Moreover, because the V3–V4 region resolves the microbiota reliably at the genus but not the species level, individual probiotic species such as *L. rhamnosus* could not be distinguished; at the genus level, the relative abundance of *Lactobacillus* did not differ markedly among treatments, consistent with the encapsulated strains constituting a minor fraction of the total community that acts mainly by modulating the resident microbiota and its metabolic output.

In contrast to alpha diversity, beta diversity analysis revealed a clear difference between the experimental groups. With ANOSIM, there were statistically significant differences between the groups ($R = 0.463$, p -value = 0.002; Fig. 3C), and with NMDS analysis, a well-defined ordination of AET3 and AET8 in contrast to AEN and AEP groups was obtained with low stress (0.02) (Fig. 3D). Taxon-level PCA accounted for

87.56% of variation, with PC1 responsible for 72.08% and PC2–15.48%. Similarly, the four groups were separated with PCA based on OTUs (Fig. 3E). PLS-DA revealed four distinct clusters of groups, with AET3 and AET8 clearly being distinguished from other groups (Fig. 3F). Bray-Curtis PCoA analysis was shown in Supplementary data (Fig. S3A). Additional taxonomic resolution was provided by the phylum-level barplots, species phylogenetic analysis, and LEfSe cladogram, which further illustrated treatment-dependent shifts in microbial composition and discriminatory taxa (Fig. S4A–C). Thus, it can be stated that Alg:PSE microbeads were able to induce the reproducible alteration of the structure of microbial community despite the lack of change in total richness.

Such trends were previously reported in case of synbiotics and polysaccharide systems, where shifts in community structure (beta diversity) were pronounced while changes in community diversity (alpha diversity) were less expressed [36,37]. Additionally, our own studies with date seed and date pomace polysaccharides indicated that such polymers led to community restructuring towards saccharolytic bacteria and SCFAs-producers [16,17,38,39].

3.5.3. Correlation between microbial function and taxa

PICRUSt2 prediction of functional capacity in each group exhibited a largely similar repertoire of metabolic capabilities within all groups, but consistent re-weighting in the microbead treatments (Fig. 4A–E). At KEGG level 2, metabolic processes were prevalent in all groups, with particular predominance in amino acid, carbohydrate, lipid, energy, cofactor/vitamin, and membrane transport metabolism (Fig. 4A,B). Based on the functional heatmap, more similarity could be seen between AEN and AEP groups, while a shift towards different functional capabilities was seen in AET3 and AET8 groups in accordance with the beta diversity analysis performed earlier.

This tendency was apparent in the analysis of COG profiles as well, which continued to demonstrate dominance of general housekeeping

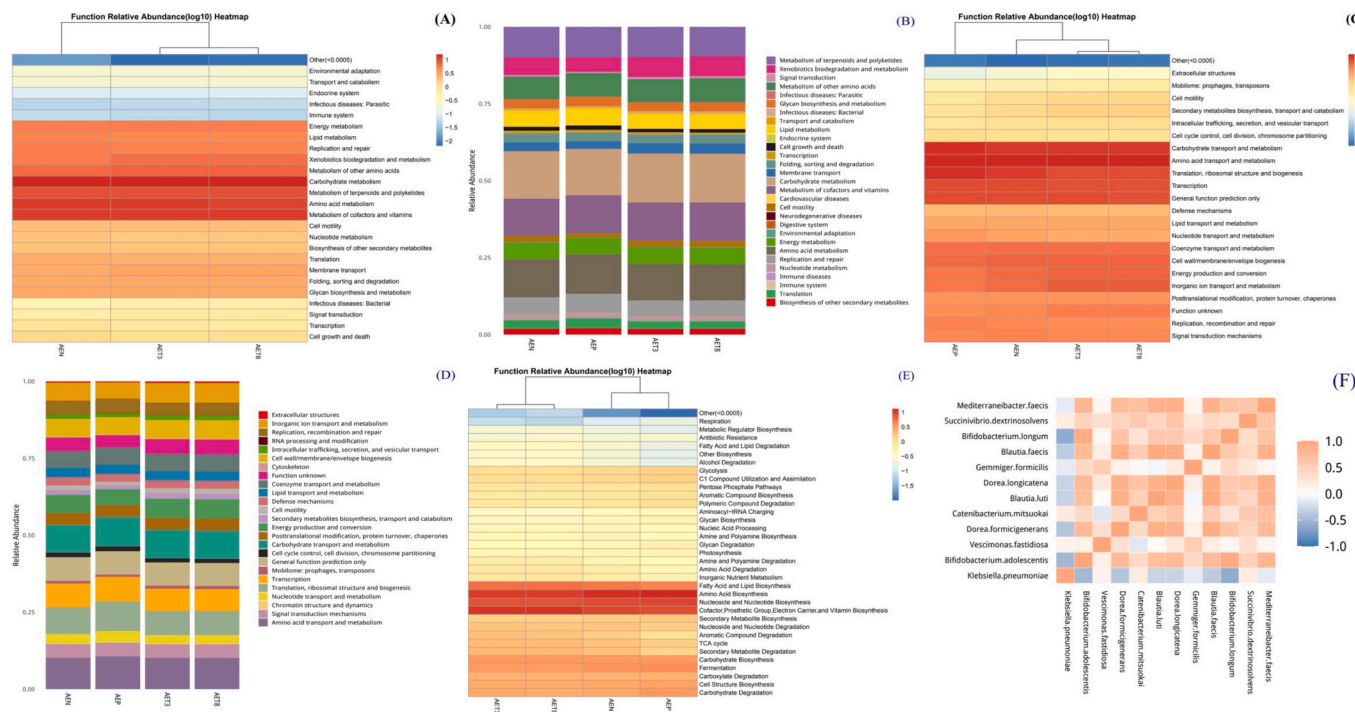


Fig. 4. Predicted microbial functional profiles and species correlations after 24 h in vitro fecal fermentation. (A) KEGG Orthology (KO) level-2 heatmap showing relative abundance of predicted microbial functions. (B) KO level-2 stacked bar plot showing the distribution of predicted KEGG functional categories across AEN, AEP, AET3, and AET8. (C) COG level-2 heatmap of predicted microbial functional categories. (D) COG level-2 stacked bar plot showing the relative abundance of predicted COG functions among treatments. (E) MetaCyc level-2 heatmap showing treatment-related variation in predicted metabolic pathways. (F) Spearman correlation heatmap showing associations among selected bacterial species enriched during fecal fermentation. AEN, negative control; AEP, GOS-P positive control; AET3, *L. rhamnosus* encapsulated in alginate + PSE (1:2, v/v); AET8, *B. longum* subsp. *longum* encapsulated in alginate + PSE (1:2, v/v).

and metabolic functions such as amino acid transport and metabolism, signal transduction mechanisms, nucleotide transport and metabolism, energy production and conversion, transcription, and translation. As a result, neither AET3 nor AET8 formed completely novel functional capabilities within the community, rather shifted the re-weighting within existing microbiome functions. Similarly, based on MetaCyc functional annotation, the dendrogram of AET3 and AET8 showed higher prevalence of carbohydrate degradation, carboxylate degradation, fermentation, metabolism associated with TCA cycle, amino acid biosynthesis, and fatty acid/lipid metabolism than those in AEN and AEP (Fig. 4E), which corresponds well with early peaks of propionate and butyrate found in Table 2.

Taxonomic correlation structure was no less interesting, too. Based on Spearman correlation heatmap, there were many positive associations within *Bifidobacterium longum*, *Bifidobacterium adolescentis*, *Blautia luti*, *Blautia faecis*, *Dorea longicatena*, *Dorea formicigenerans*, *Gemmiger formicilis*, *Mediterraneibacter faecis*, *Catenibacterium mitsuokai*, and *Succinivibrio dextrinosolvens*, while *Klebsiella pneumoniae* had a negative correlation with these taxa (Fig. 4F). Similarly, the species network analysis demonstrated the same trends (Fig. S3B). Biologically, the identified species represent typical fermenting saccharolytics and producers of SCFAs (*Blautia*, *Dorea*, *Gemmiger*, *Bifidobacterium*). *Klebsiella pneumoniae*, however, is often considered to be an opportunistic pathobiont [43]. Negative correlation of *Klebsiella* and beneficial fermenters is also known in other synbiotic studies [44,45].

3.6. Untargeted metabolite profiling of fecal fermentations

Approximately 15,000 features were identified using untargeted LC-MS across all four treatments, AEN, AEP, AET3, and AET8. After applying statistical filters, the main discriminant features were reported in Supplementary Tables S3 and S4. Some examples include organic acids, lactic acid-derived features, acetaldehyde derivatives, citrulline, ornithine, ammonia, and arginine-related features, as well as various lipids-like ions. Examples of repeat discriminant features included m/z 208.1334 (RT 5.16 min), 141.0659 (RT 19.39 min), 240.1803 (RT 5.23 min), 275.1103 (RT 5.37 min), and the high-mass ion at m/z 663.4567 (RT 12.93 min), which appeared to be cardiolipin/phospholipid-like.

PCA showed a clear clustering of the probiotic-encapsulated samples from the controls (Fig. 5A). The first two components explained 58.82% and 18.40% of total variance. PC1 separated AET3 most clearly, suggesting that this sample had the greatest metabolome-wide shift, while PC2 separated AET8. This demonstrates that both microbeads were able to induce significant changes in the fecal metabolome, albeit partially strain-specific.

Metabolite-genus associations for AET3 and AET8 samples (Fig. 5B and C) revealed interesting mechanistic information about these metabolic shifts. For example, *Bifidobacterium*, *Bacteroides*, *Blautia*, and *Dorea* were associated with several metabolites belonging to amino acids, nucleotides, carbohydrates, and SCFAs and likely reflected active saccharolytic fermentation in the sample. At the same time, *Klebsiella* and *Escherichia* showed a distinctly different pattern of association. The genus-feature heatmap (Fig. 5C) further showed that many of the discriminant RT- m/z features for AET3 and AET8 corresponded with the

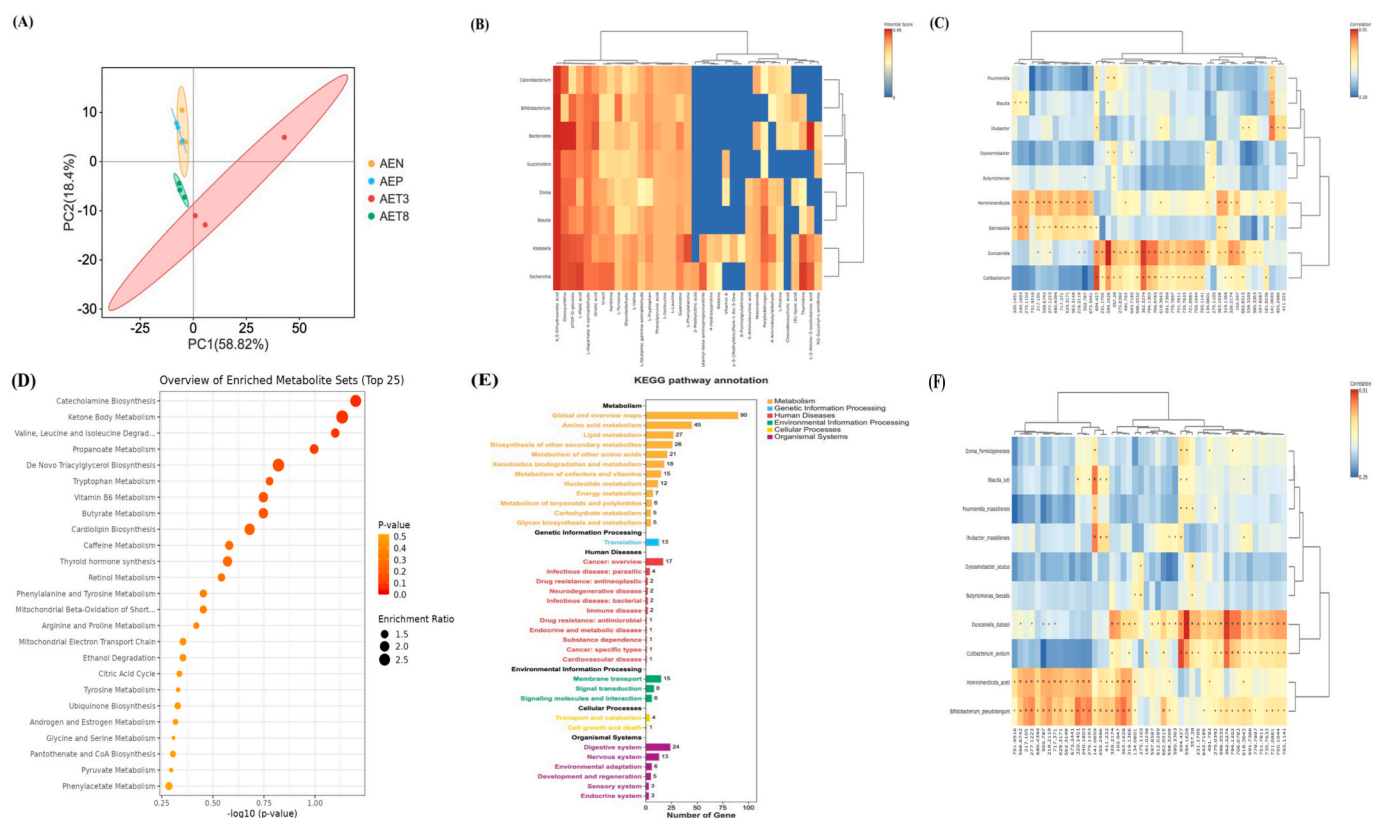


Fig. 5. Untargeted metabolomic profiling and pathway analysis of fecal fermentation supernatants after 24 h. (A) Principal component analysis (PCA) score plot showing separation of AEN, AEP, AET3, and AET8 based on LC-MS metabolite features. (B) Genus-compound heatmap showing associations between selected bacterial genera and annotated discriminatory metabolites. (C) Genus-feature heatmap showing correlations between selected bacterial genera and major discriminatory LC-MS features (RT- m/z); asterisks indicate significant associations. (D) Bubble plot showing the top enriched metabolite sets; bubble size represents enrichment ratio and color represents P-value. (E) KEGG pathway annotation of discriminatory metabolites detected in AEN, AEP, AET3, and AET8 after 24 h fecal fermentation. (F) Spearman correlation heatmap showing associations between selected bacterial species. AEN, negative control; AEP, GOS-P positive control; AET3, *L. rhamnosus* encapsulated in alginate + PSE (1:2, v/v); AET8, *B. longum* subsp. *longum* encapsulated in alginate + PSE (1:2, v/v).

presence of *Barnesiella*, *Hominimerdicola*, *Butyricimonas*, *Blautia*, and *Thubacter*, while genera such as *Cutibacterium* exhibited weak or inverse correlations with those metabolites. These findings are consistent with well-known roles of *Blautia*, *Dorea*, and *Butyricimonas* in amino acid fermentation and production of SCFAs [40]. These correlations are also in agreement with an increased early butyrate and propionate content described in Table 2.

Enrichment analysis of metabolite sets showed that the most frequently enriched pathways included catecholamine biosynthesis, ketone body metabolism, valine/leucine/isoleucine degradation, propanoate metabolism, tryptophan metabolism, vitamin B6 metabolism, butyrate metabolism, and cardiolipin biosynthesis (Fig. 5D). This shows that apart from promoting carbohydrate fermentation, the probiotics also influenced lipid, amino acid, and signaling pathways. Enrichment of propanoate and butyrate metabolism was expected based on our earlier finding of increased butyrate content in Table 2. Additionally, enrichment of cardiolipin and other lipid metabolism pathways is consistent with the metabolic remodeling occurring during bacterial membrane synthesis during fermentation [41,42]. In particular, a repeated observation of m/z 663.4567 ion as one of the discriminant features strongly supports this hypothesis.

KEGG annotation of the differential metabolites showed that most of the identified features belonged to general metabolic classes (Fig. 5E). The largest number of metabolites was found in global and overview maps (90), followed by amino acid metabolism (45), lipid metabolism (27), biosynthesis of other secondary metabolites (26), metabolism of other amino acids (21), xenobiotic biodegradation and metabolism (18), metabolism of cofactors and vitamins (15), and nucleotide metabolism (12). Other small annotations were assigned to carbohydrate and glycan metabolism (5 each), as well as some host-focused categories such as digestive system (24) and nervous system (13). Overall, this information suggests that while both GOS-P and the Alg:PSE microbeads induced SCFA metabolism, AET3 and AET8 favored increased emphasis on lipid and signal-related metabolic activity. The variable-importance heatmap further supported treatment-dependent separation among AEN, AEP, AET3, and AET8 (Fig. 5F).

In summary, untargeted metabolomics confirmed that probiotic loading inside the Alg:PSE2 matrix not only preserved their viability but also altered their fermentation behavior. Metabolites produced during probiotic-mediated fermentation within the matrix were distinct from both negative control and positive control (freely fermentable GOS-P control). The strongest alterations were seen in AET3 and AET8, with the first treatment having the largest effect overall. These results are in agreement with SCFA and metagenomic observations and suggest that our Alg:PSE2-based delivery system can be considered a potential synbiotic candidate.

4. Conclusion

This investigation confirmed that incorporating a UPS-stabilized pickering emulsion into alginate microbeads could significantly enhance probiotics' delivery properties. Among all the combinations analyzed, Alg:PSE2 was selected as the optimal system due to the superior combination of probiotics' entrapment efficiency and gastrointestinal resistance for both *Lactobacillus rhamnosus* (AET3) and *Bifidobacterium longum* subsp. *longum* (AET8). The optimized microbead formulation yielded a tougher, less swellable, and heat-resistant matrix compared with control beads without emulsifier and led to better viability during simulated digestion. Moreover, the presence of UPS-stabilized pickering emulsions could promote the antioxidant activity and the capability of inhibiting digestive enzymes of microbeads' bioavailable fractions.

In an in vitro fecal fermentation model, AET3 and AET8 exhibited an earlier response during fermentation compared with the corresponding control beads based on elevated propionate and butyrate contents at early incubation periods. Microbial community structure and predicted

functional profiles showed that probiotics significantly enriched the SCFAs-related genera (e.g., *Blautia*, *Dorea*, *Gemmiger*, and *Bifidobacterium*), with decreased abundance of pathogens (e.g., *Klebsiella*). Furthermore, the microbead matrix contributed to a unique metabolomic profile including SCFAs, amino acids, and lipids, indicating that the alginate/pickering emulsion system impacted probiotic bacteria not only through promoting their survival but also via facilitating metabolic products. More interestingly, the effects caused by the microbeads partially overlapped with the response to the prebiotic GOS-P, suggesting the unique effect of Alg:PSE2 matrix on shaping gut microbiome in vitro.

In conclusion, this research indicated Alg:PSE microbeads to be promising potential synbiotic candidates with excellent properties for probiotic colonic release and targeted gut microbiota modulation. However, it is noteworthy that this research only involved in vitro simulated human gastrointestinal digestion and in vitro fecal fermentation and metabolites obtained from untargeted liquid chromatography–mass spectrometry were putatively annotated based on known chemical databases. Thus, future work should focus on evaluating the impact of Alg:PSE2 microbeads on probiotic stability during storage, release kinetics, and in vivo performance.

A particular strength of this study is its integrated design, which links matrix characterization with probiotic survival, post-digestion bioactivity, SCFA production, 16S rRNA microbiota profiling, and untargeted metabolomics for two distinct probiotic strains within a single date seed-derived synbiotic platform. Nevertheless, several limitations should be acknowledged. First, the in vitro fecal fermentation did not include an empty Alg:PSE2 bead control (matrix without probiotics), so the relative contributions of the matrix and of the released probiotics cannot be fully separated. Second, texture analysis was performed on highly hydrated beads (moisture 90–97%), so the reported values characterize the hydrated system. Third, the pickering nature of the emulsion was inferred from the particulate properties of UPS rather than from direct interfacial measurements. Fourth, lactic acid was not quantified in the SCFA assay, although it is a major primary metabolite of *L. rhamnosus* and a precursor of butyrate and propionate. Fifth, metabolite identifications are putative (MSI level 2–3). Finally, strain-specific stimulation was not quantified; accordingly, the system is described as a potential synbiotic candidate, and confirmation of selectivity (e.g., by strain-specific qPCR), together with storage, release-kinetics, and in vivo studies, remains an objective for future work.

Abbreviations

PSE	Date seed Polysaccharide based emulsion
UPS	Ultrasound-extracted date seed polysaccharide
DSC	Differential Scanning Calorimetry
TGA	Thermo-gravimetric analysis
SEM	Scanning Electron Microscopy
FTIR	Fourier Transform Infrared Spectroscopy
DPPH	1-diphenyl-2-picrylhydrazyl
ABTS	2,2'-asino-bis(3-ethylbenzene-thiazoline-6-sulphonic acid
SOD	Superoxide Dismutase
SAS	Superoxide Anion Radical Scavenging
FRAP	Ferric Reducing Antioxidant Power
TAC	Total Antioxidant Capacity
ACE	Angiotensin Converting Enzyme
COG	Clusters of Orthologous Groups
FLASH	Fast Length Adjustment of Short
GOS-P	Galacto-oligosaccharide-Positive Control.
KEGG	Kyoto Encyclopedia of Genes and Genomes
LEfSe	Linear Discriminant Analysis Effect Size
LSPR	Localized Surface Plasmon Resonance
NC	Negative Control
OTU	Operational Taxonomic Unit
PCA	Principal Component Analysis

PICRUST Phylogenetic Investigation of Communities by Reconstruction of Unobserved States
 PLS-DA Partial Least Squares Discriminant Analysis
 PS – Polysaccharides.
 SCFAs Short-Chain Fatty Acids
 UPGMA Unweighted Pair Group Method with Arithmetic Mean

CRedit authorship contribution statement

Athira Jayasree Subhash: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Gafar Bamigbade Babatunde:** Writing – review & editing, Methodology, Investigation. **Maduni Jayahansi Paththuwe Arachchi:** Writing – review & editing, Validation, Methodology. **Abdelmoneim Ali:** Writing – review & editing, Validation, Resources, Methodology. **Afaf Kamal-Eldin:** Writing – review & editing, Validation, Resources. **Mutamed Ayyash:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Informed consent statement

Informed consent was obtained from human subjects involved in the study. The protocol for fecal sample collection was approved by the UAEU-Human Research Ethics Committee (UAEU-HREC) of United Arab Emirates University, UAE (No. ERH_2024_5441).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors express their gratitude to the United Arab Emirates University for funding this project (grant number #12F049).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2026.153313>.

Data availability

Data will be made available on request.

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